Sharp thresholds for five-colour cube assemblies

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Abstract

We determine the least pool sizes guaranteeing a frame or a complete $a \times b \times c$ box from unit cubes whose faces use all five colours of a fixed palette, for every $a, b, c \geq 2$. Proper rotations are allowed; the target has monochromatic exterior faces using all five colours. The frame threshold is $\max\{15, 4(a+b+c)-16, 2(a+b+c)+1\}$, and the complete-box threshold is $\max\{abc, 15, 2(a+b+c)+1\}$. Thus the frame threshold depends only on the sum of the side lengths, and only three complete-box shapes can require spare sources. The proof combines a certified fifteen-cube corner theorem with elementary Hall arguments and a uniform three-variety obstruction. It completes the five-colour frame column left open by Berkove, Eickemeyer and Schulman (2018). Lean verifies the entire classification. A fixed-size flow formulation constructs assemblies in time linear in the explicit input and output sizes. We also determine the corner thresholds when the number of source varieties is bounded.

AI transparency. The research, code and manuscript were developed using GPT-6-Astra (OpenAI), through Codex, under the author's direction. This working manuscript has not received independent human peer review; Section 10 states the formal verification scope.

1 Introduction

MacMahon's coloured-cube puzzles [1] ask when a collection of face-coloured cubes can form a larger cube with monochromatic exterior faces. Here every source and target uses all five colours of a fixed palette. Proper rotations identify varieties; reflections are not allowed as moves. Collections may contain repeated varieties, and touching interior faces need not have equal colours. For $a, b, c \geq 2$, a *frame* consists of the eight corners and the noncorner positions on the twelve edges of an $a \times b \times c$ box. It has $L = 4(a+b+c)-16$ positions. Write $F_5(a,b,c)$ for the least pool size guaranteeing a frame and $G_5(a,b,c)$ for the least pool size guaranteeing a complete box. An assembly may use a subset of the pool; its target colouring may depend on the collection.

Theorem 1 (All cuboid thresholds)**.** *For every integer $a, b, c \geq 2$,*

$$F_5(a,b,c) = \max\{15, 4(a+b+c)-16, 2(a+b+c)+1\},$$
$$G_5(a,b,c) = \max\{abc, 15, 2(a+b+c)+1\}.$$

The frame threshold depends only on the sum $s = a+b+c$: it is 15 for $s = 6, 7$, 17 for $s = 8$, and the frame's own cardinality $4s-16$ for $s \geq 9$. Up to permuting sides, only three complete boxes can require spare sources: $2\times2\times2$, $2\times2\times3$, and $2\times2\times4$, whose thresholds are $15, 15, 17$. The distinction between the two thresholds matters: a $2\times3\times3$ frame has sixteen positions and needs seventeen available sources, whereas its complete box uses any eighteen sources without spares. Table 1 lists the smallest shapes; from $s = 9$ on, both thresholds equal the number of positions.

For ordinary cubes, abbreviate $F_5(n) = F_5(n,n,n)$ and $G_5(n) = G_5(n,n,n)$. The classification immediately gives the following.

Box	s	Frame positions	F_5	Volume	G_5
$2\times2\times2$	6	8	**15**	8	**15**
$2\times2\times3$	7	12	**15**	12	**15**
$2\times2\times4$	8	16	**17**	16	**17**
$2\times3\times3$	8	16	**17**	18	18
$2\times2\times5$	9	20	20	20	20
$2\times3\times4$	9	20	20	24	24
$3\times3\times3$	9	20	20	27	27

Table 1: Thresholds for the smallest boxes. A bold entry exceeds the number of positions: some collections with no spare sources fail.

Corollary 2 (Ordinary cubes)**.**

$$F_5(n)=\begin{cases}15, & n=2,\\ 12n-16, & n\geq 3,\end{cases}\qquad G_5(n)=\max\{n^3,15\}\quad (n\geq 2).$$

In particular, any twenty sources form the frame of a $3\times3\times3$ *cube, and any 27 form a complete* $3\times3\times3$ *cube.*

Berkove, Eickemeyer and Schulman [2, Section 8, Table 7] record frame thresholds for palettes of two to six colours, leaving the five-colour column open. Table 2 inserts the values proved here; the cuboid classification goes beyond that table.

Side length	2 colours	3 colours	4 colours	**5 colours**	6 colours
$n=2$	11	23	11	**15**	24
$n=3$	20	35	20	**20**	24
$n\geq4$	$12n-16$	$16n-17$	$12n-16$	$\mathbf{12n-16}$	$12n-16$

Table 2: Cube-frame thresholds. The five-colour column is proved here; other columns are as recorded in [2, Table 7].

We first prove the corner threshold independently of the extensions. Write $N_5=F_5(2)=G_5(2)$.

Theorem 3 (Corner threshold)**.** $N_5=15$.

An elementary fourteen-cube obstruction proves the lower bound. For the upper bound, Hall deficiencies and symmetry reduce hypothetical counterexamples to finite problems, excluded by interval certificates (Sections 4–5). A prescribed-family corner lemma, proved independently of $N_5=15$, and geometric compatibility bounds then extend this input to every cuboid (Sections 6–7). A uniform three-variety obstruction supplies the additional lower bound responsible for the two sixteen-position exceptions. These extensions require no new interval certificates. Section 8 gives a compressed flow algorithm for finding assemblies, and Section 9 determines the thresholds for collections with a bounded number of varieties. The threshold classification is fully formalised in Lean; Section 10 describes the verification scope.

The side-length restriction is substantive. With a side of length one, a source can meet two opposite exterior faces and expose four or more faces at once. The corner-and-edge model and the completion by arbitrary unused cubes then need replacement.

The accompanying repository contains the certificates, programs, geometric audits and Lean sources.[1]

[1] https://github.com/jasondavies/five-colour-cubes. A fixed review archive includes source checksums, the base Git revision and bundled pinned project dependencies.

The companion paper [5] considers four-colour sources and targets using exactly three colours. It gives the sharp frame threshold $16n - 3$ for every side length $n \geq 2$, with a classification of the largest failing collections. The two developments share the palette-independent geometry in the `colour-cubes-core` library [6]; neither threshold theorem is used to prove the other.

A limited literature search has found no later value for the five-colour entry. This is not a comprehensive priority claim. In particular, Rosenfeld's October 2026 article [7] has not been examined in full.

2 Geometry and Hall deficiencies

Represent a cube by its colours on the faces $(+x, -x, +y, -y, +z, -z)$ in a right-handed coordinate frame. A five-colour cube has one repeated colour. There are 75 varieties: for each repeated colour, three have its two faces opposite and twelve have them adjacent. In the opposite case, the other four colours form two opposite pairs, giving three possibilities. In the adjacent case, fix the two equal faces; the 24 orders of the other colours are identified in pairs by the rotation exchanging those equal faces. Thus there are 15 opposite-repeat varieties and 60 adjacent-repeat varieties. Each class is one orbit under colour permutations.

A corner type is the ordered triple of its incident face colours, modulo cyclic permutation. Order the outward face normals positively, equivalently read the triple counterclockwise from outside along the body diagonal. This reverses the clockwise convention in [2], consistently on every cube, and therefore preserves all matching and containment relations. Repeated occurrences of a corner type are retained as distinct target positions.

Lemma 4 (Corner–rotation correspondence)**.** *A source cube fits a target corner by a proper rotation if and only if it has the same cyclic corner type at some corner.*

Proof. A proper rotation preserves the orientation of a corner frame, so it induces a cyclic permutation of its three outward faces. Conversely, any two positively oriented corner frames, with a prescribed cyclic alignment, are related by a determinant-positive signed permutation of the coordinate axes. This is a proper rotation of the cube. □

The palette-independent Lean version of this lemma is reused from the shared `colour-cubes-core` library [6].

Fix a target T. Form a bipartite graph whose left vertices are its eight corner positions and whose right vertices are the individual source copies. A copy is adjacent to a position when it fits there. A matching covering the eight positions is exactly a solution: each chosen copy can be rotated independently, since interior faces are unrestricted. By Hall's theorem [3], failure is equivalent to a nonempty set D of target positions having fewer than $|D|$ neighbouring copies.

Index the varieties by $i = 1, \ldots, 75$, and let x_i be their multiplicities. For a set S of varieties, write $x(S) = \sum_{i \in S} x_i$. Let $N_T(D)$ be the varieties fitting at least one position of D. Then failure of T is equivalent to the existence of a cut

$$x(N_T(D)) \leq |D| - 1, \qquad \varnothing \neq D \subseteq \{1, \ldots, 8\}. \tag{1}$$

There are 255 cuts per target, hence 19,125 cuts altogether.

3 An elementary fourteen-cube obstruction

Consider four copies of A, five of B, and five of C, where

$$A = (0, 0, 1, 2, 3, 4), \quad B = (0, 1, 2, 3, 2, 4), \quad C = (0, 3, 1, 2, 4, 1).$$

Their corner types are shown below. Triples denote cyclic classes; the last column counts positions whose type occurs in neither of the other varieties.

Variety	Corner types	Private positions
A	$013, 014, 023, 024, 031, 032, 041, 042$	5
B	$022, 023, 034, 042, 122, 124, 132, 143$	6
C	$011, 014, 021, 042, 113, 123, 134, 243$	6

Lemma 5. *The collection $4A + 5B + 5C$ has no five-colour solution.*

Proof. A solvable target must have every corner type in the displayed union. Its only two-colour types are $022, 122, 011, 113$. An adjacent-repeat target therefore repeats either colour 2, with types $022, 122$, or colour 1, with types $011, 113$. In either case these two corners determine the colours on the faces at the ends of the repeated-colour edge. The two orders of the remaining colours give a mirror pair: B and its mirror, or C and its mirror. The mirror of B requires 043, and the mirror of C requires 012; both types are absent. Only B, C remain.

An opposite-repeat target requires both cyclic orientations of each corner triple. The unordered triples with both orientations in the union are

$$013, 014, 023, 024, 123, 134.$$

If its repeated colour is r, delete r from the four triples containing it. The remaining pairs must form a $K_{2,2}$ on the other four colours: choose one face from each of their two opposite pairs. Colours 2 and 4 occur in fewer than four listed triples. For $r = 1$ or $r = 3$, the four pairs have degree sequence $(3, 2, 2, 1)$, so do not form $K_{2,2}$. For $r = 0$, the pairs are $13, 14, 23, 24$, forcing opposite pairs $\{1, 2\}$ and $\{3, 4\}$. This is A.

Thus only A, B, C can be targets. Their private positions require respectively at least five, six and six copies of the corresponding variety, exceeding the available four, five and five copies (Figure 1). None is solvable. □

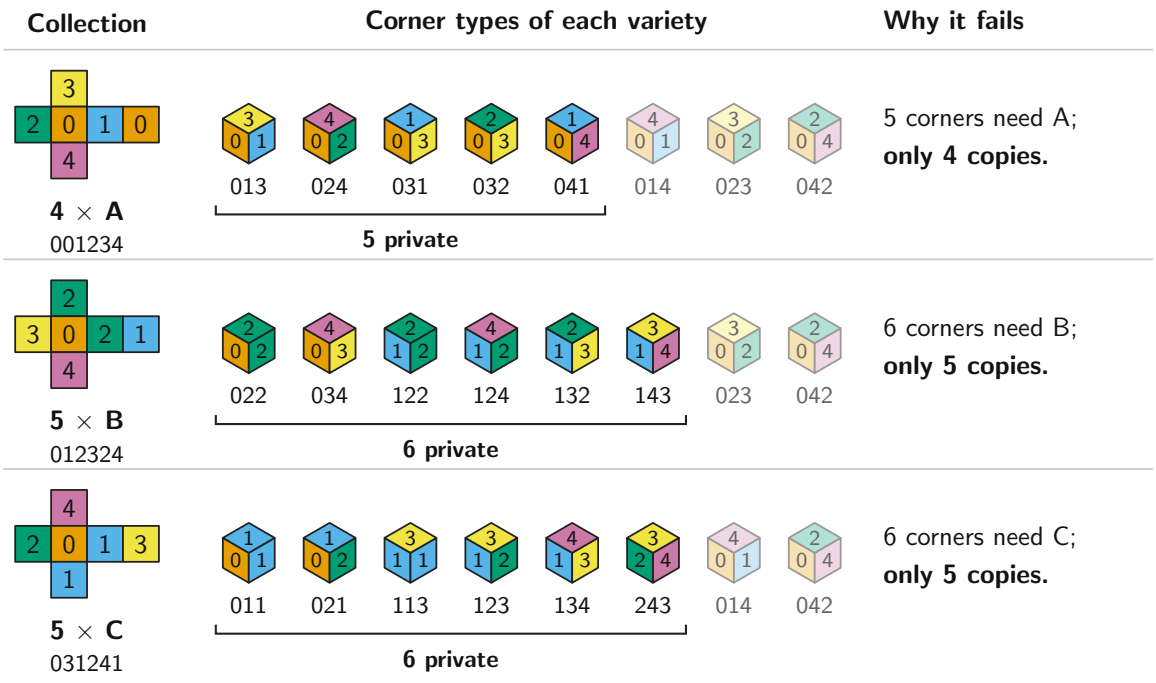


Figure 1: The fourteen-cube obstruction $4A + 5B + 5C$. Each net has $+x$ at its centre. Corner types are read counterclockwise from outside; the full-colour types occur on no other variety. A target of variety X has these private positions, and only copies of X fit them.

4 Reduction to finite problems

Eight copies of any variety solve that variety, so a solution-free collection has every multiplicity at most seven. Suppose such a collection has size 15.

Lemma 6 (Anchored Hall cover)**.** *Suppose a collection contains m copies of a variety A and does not solve A. There is a set K of its target positions such that*

$$|K| \le 7 - m, \qquad \#\{\text{other copies fitting some position outside } K\} \le 7 - m - |K|. \tag{2}$$

Conversely, such a set certifies failure of target A.

Proof. Apply (1) to A and put $K = D^c$. Every copy of A is a neighbour of the nonempty set D. Subtract these m copies from the bound $|D| - 1 = 7 - |K|$. The converse is the same Hall inequality read backwards. □

First suppose the maximum multiplicity m is at least two. Choose an adjacent-repeat anchor attaining m if possible; otherwise choose an opposite-repeat anchor attaining m. In the latter case all adjacent-repeat multiplicities are at most $m-1$. Relabel the colours to send the anchor to a fixed representative of its orbit.

For each cover K in Lemma 6, put

$$S(K) = N_A(K^c) \setminus \{A\}, \qquad b(K) = 7 - m - |K|.$$

Write $S = S(K)$ and $b = b(K)$. The case has $x_A = m$, $x(S) \le b$, and the multiplicity bounds just described. Additional upper bounds use two-variety solutions. If j copies of A and $8 - j$ copies of V build a target, with $j \le m$, then $x_V \le 7 - j$ in a solution-free collection. These tests are finite: the two varieties must cover all eight target positions, A must fit at least j, and V at least $8 - j$. They are also sufficient: at most j positions are unavailable to V, and all of them are available to A. Assign these to A, extend the assignment to j positions that A fits, and assign the remaining $8 - j$ positions to V.

Colour permutations and simultaneous reflection of the entire collection preserve solvability. First identify equal pairs $(S(K), b(K))$, even when they arise from inequivalent position subsets K; then retain one representative per orbit of these pairs under the anchor stabiliser. The counts below refer to these constraint pairs. Transporting a cover also transports its constrained varieties, budget and multiplicities, so no counterexample is lost. Global reflection transports the whole collection; individual cubes still move only by proper rotations. The repeated-variety cases have the following counts, totalling 244.

Anchor / maximum multiplicity	2	3	4	5	6	7
Adjacent-repeat	72	55	31	14	4	1
Opposite-repeat	26	21	11	6	2	1
Total	98	76	42	20	6	2

For distinct collections a different choice of anchor reduces the count.

Lemma 7 (Shared two-colour corner)**.** *Among eleven adjacent-repeat cubes, two share a two-colour corner type. If a distinct collection contains such a pair and one member is chosen as anchor, neither two-colour position of the anchor can be Hall-deficient.*

Proof. Each adjacent-repeat cube with repeated colour r has exactly two such types, rrx and rry, with $x \ne y$. There are only $5 \cdot 4 = 20$ types rrx. Eleven cubes contribute 22 incidences, so two cubes share a type. For the two characteristic positions of the chosen anchor, the anchor itself and the other cube are two distinct neighbours. Their neighbourhood therefore has size at least two. □

If an opposite-repeat variety is present, choose it as anchor; this gives 31 constraint orbits with every multiplicity at most one. Otherwise use Lemma 7 to choose an adjacent-repeat anchor. Omit the cover complementary to its two characteristic positions and set all opposite-repeat capacities to zero. Its stabiliser preserves these positions, so the omission is compatible with the orbit reduction. It removes one of the 83 adjacent-repeat constraint orbits, leaving 82 further cases.

Proposition 8. *Every solution-free fifteen-cube collection is carried by colour relabelling and, if needed, global reflection into one of the 357 exported root cases.*

Proof. The preceding arguments cover repeated and distinct collections. The finite catalogue, two-variety bounds and coverage of all anchor-cover orbits are checked from physical rotations by the accompanying audit. It regenerates rotations from quarter turns, separately from the signed-axis permutations used by the case generator, and checks every exported bound and every cover orbit. The resulting counts are

$$244 + 31 + 82 = 357.$$

The cases need not be disjoint; completeness of the cover is what is required. □

Each root specifies an anchor A of multiplicity m, coordinate bounds ℓ, u, a set S of non-anchor varieties and a budget b. The bounds fix $\ell_A = u_A = m$, with $\ell_i = 0$ for $i \neq A$. A vector $x \in \mathbb{Z}^{75}_{\geq 0}$ is *admissible* for this root if

$$\ell \leq x \leq u, \qquad \sum_{i=1}^{75} x_i = 15, \qquad x(S) \leq b, \tag{3}$$
$$\text{for every target } T, \text{ there is } \varnothing \neq D \subseteq \{1, \ldots, 8\} \text{ with } x(N_T(D)) \leq |D| - 1.$$

Thus admissibility includes failure of every target, in addition to the anchor constraints.

The cover can be reduced without further search. If $S' \subseteq S$ and $b' \geq b$, then $x(S) \leq b$ implies $x(S') \leq b'$ for nonnegative x. Together with coordinatewise inclusion of the root boxes, this identifies 49 cases contained in other cases, leaving 308. Write u for the root upper bounds. In 29 retained cases, $b + u(S^c) < 15$ already contradicts the prescribed total, so only 279 cases need further exclusion proofs. The exact map and retained case numbers are in `data/q5-root-reduction.json`; Lean checks all containments and arithmetic exclusions. The archive retains all 357 original cases.

Lemma 9 (Hall-budget projection)**.** *Suppose a root has anchor multiplicity m, budget $x(S) \leq b$, and total 15, with the anchor outside S. For $0 \leq \lambda \leq b$, put $G = 15 - b + \lambda$ and suppose $G \geq m$. Every admissible vector x has a vector $y \leq x$ with the same anchor multiplicity, total G, budget $y(S) \leq \lambda$, and a Hall deficiency for every target. It obeys the original upper bounds, tightened to $\min(u_i, \lambda)$ on S.*

Proof. First retain only $\min(x(S), \lambda)$ copies in S. This removes at most $b - \lambda$ copies and leaves the anchor untouched, so the remaining total is at least G. Delete further non-anchor copies until the total is G; this is possible because $G \geq m$. Every coordinate decreases, so all upper bounds and Hall deficiency inequalities are preserved. Nonnegativity and $y(S) \leq \lambda$ give $y_i \leq \lambda$ for $i \in S$. □

For our anchored covers, $b = 7 - m - |K|$, so

$$G = m + 8 + |K| + \lambda.$$

Thus the reduced collection consists of the m fixed anchor copies and $8 + |K| + \lambda$ other copies; in particular, $G \geq m$ is automatic. Apply the lemma after selecting and normalising the original

root, retaining its bounds. No new anchor selection is required for the smaller collection. Of the 279 remaining roots, 277 admit an exclusion certificate with $\lambda = 0$. The other two, original cases 68 and 69, use $\lambda = 1$ and $G = 12$. Their $\lambda = 0$, $G = 11$ problems actually have solution-free collections, supplied and checked by physical matching in the projected manifest. The projected totals range from 9 to 15. All 101 roots whose original budget was zero retain their original certificates.

5 Interval certificates

For a projected problem, admissibility is defined by (3) with tightened upper bounds and total G. In this section denote its projected budget by b. We certify that no admissible vector exists.

Write a box as $\ell \le x \le u$. For a cut $x(S) \le r$, two necessary conditions for feasibility in the box are

$$\ell(S) \le r, \qquad G \le r + u(S^c). \tag{4}$$

The first follows from the lower bounds; the second follows from $G = x(S) + x(S^c)$. When they hold, put $\sigma = r - \ell(S)$ and $\gamma = r + u(S^c) - G$. Every satisfying vector obeys

$$x_i \le \min(u_i, \ell_i + \sigma) \quad (i \in S), \qquad x_i \ge \max(\ell_i, u_i - \gamma) \quad (i \notin S). \tag{5}$$

For example, the upper bound follows by giving all other coordinates of S their lower bounds; the lower bound follows by giving all other coordinates of S^c their upper bounds.

For a target, retain its cuts satisfying (4). If none remain, no vector in the box can fail that target. Otherwise let I and U be the intersection and union of their neighbourhoods, and let σ_*, γ_* be the largest corresponding σ, γ. Then all vectors in the box that fail this target remain in the refined box

$$u_i' = \begin{cases} \min(u_i, \ell_i + \sigma_*) & i \in I, \\ u_i & i \notin I, \end{cases} \qquad \ell_i' = \begin{cases} \max(\ell_i, u_i - \gamma_*) & i \notin U, \\ \ell_i & i \in U. \end{cases} \tag{6}$$

Indeed, a failing vector satisfies one of the retained cuts, and its bounds in (5) imply (6). The root budget is treated as a family containing a single cut. The total $\sum_i x_i = G$ gives

$$\ell_i' = \max\left(\ell_i, G - \sum_{j \ne i} u_j\right), \qquad u_i' = \min\left(u_i, G - \sum_{j \ne i} \ell_j\right).$$

A total-bound leaf closes exactly when $\sum_i \ell_i > G$ or $\sum_i u_i < G$. A budget leaf closes when $\ell(S) > b$ or $b + u(S^c) < G$; a target leaf closes when none of its 255 cuts satisfies (4). The checker thus has 77 closing reasons: one total test, one budget test and 75 target tests. Refinements must tighten at least one bound and leave nonempty coordinate intervals; a split must satisfy $\ell_i \le v < u_i$.

A certificate is a finite tree of refinements and integer splits $x_i \le v$ or $x_i \ge v + 1$. Each leaf is excluded by total bounds or by a cut family with no surviving cut. Induction on this tree proves that its root contains no admissible vector. The separate C++ checker reads these instructions and verifies the deductions; it performs no search and does not use the search program's matching implementation.

Proposition 10 (Certificate replay)**.** *The separate checker accepts an exclusion certificate for each of the 279 projected problems.*

The projected archive contains 499,829 instructions, 95.69% fewer than the original archive. Both archives and the search computations are documented in the repository. Every archived header is checked against its exported root case. Eighty negative controls are rejected, including

all 77 possible direct closing claims against a separate $G = 11$ fixture, not a $G = 15$ root problem. This fixture is a known solution-free eleven-cube collection: it satisfies its total, fixed anchor count, coordinate bounds and zero root budget, and fails all 75 targets by physical matching. It is `data/q5-distinct-known-control.txt`; its regeneration and witness checks are in `scripts/check_five_colour_distinct.py`.

Proof of Theorem 3. Lemma 5 gives $N_5 \geq 15$. A solution-free collection of fifteen cubes would satisfy an exported root case by Proposition 8, and would supply a Hall deficiency for every target. Containment and the initial budget tests leave one of 279 roots. Lemma 9 supplies an admissible vector in its projected problem, which Proposition 10 and the sound interval deductions exclude. Hence $N_5 \leq 15$. □

6 Compatibility and missing-pair families

The frame arguments depend on how many positions a source can miss. Lemmas 12 and 13 state the shape-independent facts used in the directional bounds of Section 7. The following form of Hall's theorem isolates the matching step.

Lemma 11 (Hall criterion with a matched subset)**.** *Consider L target positions and $N \geq L$ source copies. Suppose every source misses at most d positions. If a set P_0 of positions can be matched to distinct copies, and every position outside P_0 has at least d neighbours, then all L positions can be matched.*

Proof. For a position set $D \subseteq P_0$, the given matching supplies $|D|$ neighbours. If $D \not\subseteq P_0$ and $|D| \leq d$, a position in $D \setminus P_0$ alone has at least d neighbours. If $|D| > d$, every source neighbours D, since none can miss all its positions. In each case Hall's inequality holds. □

Lemma 12 (Missing pairs and shared corners)**.** *Every five-colour variety has eight distinct cyclic corner types. An adjacent-repeat variety lacks exactly one bichromatic edge type; an opposite-repeat variety lacks exactly two, which are disjoint. Two disjoint missing pairs determine a unique opposite-repeat variety. If either of two varieties is opposite-repeat, they share at least two corner types. A target has at most one monochromatic edge; each bichromatic type occurs on at most two edges, and a doubled type contains the repeated colour. Thus two disjoint bichromatic types account for at most three geometric target edges.*

Proof. A missing bichromatic pair occupies opposite singleton faces: if a colour is repeated, a face of the other colour cannot be opposite both copies. An adjacent-repeat variety has one opposite pair of singleton faces, and an opposite-repeat variety has two. Two disjoint missing pairs specify two opposite face pairs, leaving the fifth colour on the remaining pair, which determines a unique variety.

For the shared-corner assertion, let the opposite-repeat variety repeat r, with singleton opposite pairs $\{\alpha, \beta\}$ and $\{\gamma, \delta\}$. It contains both cyclic orientations of each rxy with $x \in \{\alpha, \beta\}$ and $y \in \{\gamma, \delta\}$. In the other variety, an r-face with no adjacent r-face has at least three distinct colours around it. Their cyclic order crosses between the two pairs at least twice, giving two shared corners. If its r-faces are adjacent, the other neighbours of each form a path of three distinct colours. Each path crosses between the pairs and gives a shared corner, and these have different types. Corner counts are symmetric: a five-colour variety has eight distinct cyclic corner types, since the only repeated colour multisets occur at the two ends of an edge, in opposite cyclic orders.

Only the repeated colour can form a monochromatic edge. Two singleton faces meet along at most one edge; hence a doubled bichromatic type contains the repeated colour and occurs at most twice. Two disjoint types cannot both contain that colour, proving the final bound. □

Lemma 13 (A transverse edge forces a common corner)**.** *Let source and target both be adjacent-repeat varieties. If the source lacks the type of a target edge not parallel to the target's monochromatic edge, then it shares a corner type with the target.*

Proof. Relabel the colours and rotate the target to $T = (0, 1, 0, 2, 3, 4)$. Its monochromatic edge is parallel to z. The types occurring on edges transverse to the monochromatic edge are $03, 04, 13, 14, 23, 24$. The proper half-turn interchanging the x- and y-directions and reversing z, followed by the colour relabelling $1 \leftrightarrow 2$, $3 \leftrightarrow 4$, fixes T and pairs these types as $(03, 04)$, $(13, 24)$, and $(14, 23)$. Thus three cases suffice. For each missing type the source has its two colours on opposite faces. The other three colours occur around that axis in one of two cyclic orientations. The three adjacent-repeat choices in each orientation share six tricoloured corners. Reading those common corners gives:

Missing type	Shared corners in one orientation	Shared corners in the other orientation
03	$024, 041$	123
13	032	$041, 142$
14	$013, 123$	024

Every entry is nonempty, so the source and target share a corner. □

For a bichromatic pair $\alpha\beta$, let $\mathcal{H}_{\alpha\beta}$ be the family of varieties lacking that edge type. In every such cube, α and β are opposite singleton faces. The four faces around this axis use the other three colours, one of them twice. There are nine varieties in $\mathcal{H}_{\alpha\beta}$: three opposite-repeat varieties O_r, and six adjacent-repeat varieties A_r^+ and A_r^-, where r is the repeated colour. The signs distinguish the two cyclic orientations of the three distinct ring colours. All three A_r^+ varieties share the same six tricoloured corner types; all three A_r^- varieties share the reverse six types.

Lemma 14 (The missing-pair corner array)**.** *The eight corners of an opposite-repeat target $O_r \in \mathcal{H}_{\alpha\beta}$ form a 2×2 array with two positions per cell. Rows correspond to the two unordered ring-colour pairs involving r, and columns to their cyclic orientations. A source O_r fits every cell; the other two opposite-repeat varieties each fit one row; and the positive and negative adjacent-repeat varieties fit the corresponding columns. Consequently, every variety in $\mathcal{H}_{\alpha\beta}$ shares at least four corner types with every opposite-repeat variety in that family.*

Proof. Each cell contains one corner at each end of the $\alpha\beta$ axis. Reading the ring colours gives the stated incidences. Figure 2 shows the example $O_2 = (0, 1, 2, 2, 3, 4)$ in $\mathcal{H}_{01}$, with $+$ chosen to include 023. Relabelling the three ring colours gives the other two targets, and relabelling the palette gives every missing-pair family. Cyclic rotations identify corner labels. In particular, the two positions in each cell have identical source neighbourhoods. □

	A_r^+ sources positive column	A_r^- sources negative column
pair 23 O_3 fits this row	$023, 132$	$032, 123$
pair 24 O_4 fits this row	$042, 124$	$024, 142$

O_2 fits all four cells

Figure 2: Corner array of O_2 in the family lacking 01. Each cell contains two target positions. Here $r \in \{2, 3, 4\}$; all three adjacent-repeat sources of a given sign fit the same column.

7 Rectangular boxes

We prove Theorem 1. A prescribed-family corner lemma and a uniform lower obstruction come first, followed by the thin-box upper bounds and the remaining shapes. The only certified input is Theorem 3; the prescribed-family lemma is independent of it.

7.1 Signed corner projections

Retain the family $\mathcal{H}_{\alpha\beta}$ from Lemma 14, with its opposite-repeat varieties and their 2×2 corner array. Write $C(W)$ for the cyclic corner types of W.

Lemma 15 (Corner projections onto a missing-pair family)**.** *Fix a bichromatic pair $\alpha\beta$. Let C^+ and C^- be the two sets of six tricoloured corner types shared by the three adjacent-repeat varieties of the respective cyclic orientations in $\mathcal{H}_{\alpha\beta}$. An opposite-repeat variety in $\mathcal{H}_{\alpha\beta}$ has four corners in each set. For a source $W \notin \mathcal{H}_{\alpha\beta}$, put*

$$p(W) = |C(W) \cap C^+|, \qquad m(W) = |C(W) \cap C^-|.$$

Then

$$(p(W), m(W)) \in \{(4,0), (2,1), (1,2), (2,2), (0,4)\}.$$

Proof. Relabel the poles as $0, 1$ and the ring colours as $2, 3, 4$. Choose the positive orientation so that

$$C^+ = \{023, 034, 042, 124, 132, 143\}, \qquad C^- = \{024, 032, 043, 123, 134, 142\}.$$

Triples denote cyclic classes; face order is $(+x, -x, +y, -y, +z, -z)$. The following table gives representatives of all geometric cases outside $\mathcal{H}_{01}$, up to a colour permutation preserving the pole pair and a simultaneous reflection, which may exchange C^+ and C^-. Such a reflection is only a symmetry of this incidence calculation, not an allowed assembly move.

W	$p(W)$	$m(W)$
$(0,0,1,2,3,4)$	2	2
$(0,1,0,2,3,4)$	0	4
$(0,2,0,3,1,4)$	2	1
$(0,2,1,2,3,4)$	0	4
$(0,2,1,3,2,4)$	2	1
$(0,2,1,3,4,4)$	2	2

To see that these cases exhaust the possibilities, first suppose a pole colour is repeated. Its two faces are either opposite, or adjacent; in the latter case the other pole is opposite one repeated face or adjacent to both. If a ring colour is repeated, the two pole faces must be adjacent, since otherwise W belongs to $\mathcal{H}_{01}$. Fix them at x^+ and y^+. Among the remaining faces x^-, y^-, z^+, z^-, an opposite repeated pair is z^+, z^-. An adjacent repeated pair is either x^-, y^-, or one of x^-, y^- together with one of z^+, z^-. These are the final three rows.

Each entry follows by reading the eight corners of the displayed cube. The projections of the nine family varieties themselves are $(6,0), (0,6)$, and $(4,4)$, as follows directly from their ring descriptions. □

Lemma 16 (Compatibility for a thin frame)**.** *Choose an orientation of a five-colour target for a $2 \times 2 \times 3$ frame such that its four noncorner edge positions are bichromatic. Every source cube fits at least four of the twelve positions.*

Proof. If either source or target is opposite-repeat, Lemma 12 gives two shared corners. The four parallel bichromatic edges have either four distinct types or two types occurring twice each. In the latter case both contain the target's repeated colour. The source's missing pairs are disjoint, so it fits at least two of these edges, giving four compatible positions.

If both varieties are adjacent-repeat, the four parallel edge types are distinct and transverse to the monochromatic edge. The source fits either all four, or at least three and a corner by Lemma 13. Again it fits four positions. □

7.2 Eight family cubes and three arbitrary additions

Lemma 17 (Eight cubes with a common missing pair)**.** *Eight cubes from $\mathcal{H}_{\alpha\beta}$, together with any three additional five-colour cubes, have a corner solution whose target belongs to $\mathcal{H}_{\alpha\beta}$.*

Proof. Let O, P, Q count the opposite-repeat, positive adjacent-repeat, and negative adjacent-repeat copies among the chosen eight. Thus $O + P + Q = 8$; C^+ and C^- are as in Lemma 15.

If $O + \min(P, Q) \geq 4$, the 2×2 Hall argument gives an opposite-repeat corner solution from these eight copies alone. Explicitly, choose O_r whose opposite-repeat multiplicity is maximal. Each cell has at least $O/2 + \min(P, Q) \geq 2$ neighbours, each row has at least $8 - O/2 \geq 4$, and each column has at least $O + \min(P, Q) \geq 4$. Every other nonempty union of cells has all eight copies as neighbours.

Otherwise exchange the signs so that $O + Q \leq 3$. Put $\kappa = O + Q = 8 - P$, so $P \geq 5$ and $0 \leq \kappa \leq 3$. Some positive variety A_r^+ occurs at least twice. If its eight corners have a matching among all eleven copies, we are done. Otherwise, choose a deficient Hall set D of its corners. The two copies of A_r^+ fit every corner, so $|D| \geq 3$ and D contains a tricoloured corner. All P positive copies are therefore neighbours of D. Let R be its other neighbours and put $U = C^+ \setminus D$, $u = |U|$. Since $|D| + u \leq 8$,

$$P + |R| < |D| \leq 8 - u, \qquad |R| + u \leq \kappa - 1.$$

Remove the P positive copies and the copies in R. Every surviving copy has all its positive corners in U, and $u \leq 2$.

Every family opposite-repeat variety has four corners in C^+, so no such chosen copy remains. At least

$$\kappa - |R| \geq u + 1$$

of the chosen nonpositive copies remain, and all of them are negative adjacent-repeat copies. Each fits every corner in C^-. The remaining pool has at least

$$\kappa + 3 - |R| \geq 4 + u$$

copies.

Lemma 15 gives the following implications for every source:

$\lvert C(W) \cap C^+ \rvert \leq u$	$u = 0$	$u = 1$	$u = 2$
$\lvert C(W) \cap C^- \rvert \geq$	4	2	1.

Indeed, the family varieties have projection sizes $(6, 0), (0, 6), (4, 4)$, and every variety outside the family has one of $(4, 0), (2, 1), (1, 2), (2, 2), (0, 4)$.

Partition C^- into three pairs, indexed by the unordered pairs of ring colours. The negative half of each family opposite-repeat target is the union of two of these pairs. We will match one such four-position half using the remaining pool, then use four positive copies for its positive half.

Let j be the number of negative-family copies in the remaining pool; thus $j \geq u + 1$. If $j \geq 4$, the conclusion is immediate. If $j = 3$, another remaining copy fits at least one negative

corner; choose a target whose negative half contains it and fill the other three positions with negative-family copies.

If $j = 2$, then $u \leq 1$. Two other remaining copies each fit at least two negative corners, so they can be assigned distinct ones. Any two negative corners lie in the union of two of the three pairs. Choose the corresponding opposite-repeat target and fill its other two negative positions with the two negative-family copies.

Finally, if $j = 1$, then $u = 0$, and there are at least three other copies, each fitting at least four of the six negative corners. Choose one of these copies W. Some union of two of the three pairs contains at least three of its negative corners. In that four-position half, every other copy fits at least two positions. Hence W and any two other copies can be matched to three distinct positions: two copies have at least two neighbours, and all three together have at least three. The one negative-family copy fills the fourth position.

In every case we obtain the negative half of an opposite-repeat target in $\mathcal{H}_{\alpha\beta}$. Four of the $P \geq 5$ positive copies fill its positive half. The positive copies used here are disjoint from the surviving pool. This proves the lemma. □

7.3 A uniform lower bound

Lemma 18 (A three-variety obstruction)**.** *Let $a, b, c \geq 2$ and put $\tau = a + b + c - 6$. In face order $(+x, -x, +y, -y, +z, -z)$, define*

$$U = (0, 1, 0, 2, 3, 4), \qquad V = (0, 1, 1, 2, 3, 4), \qquad W = (0, 3, 1, 4, 2, 3).$$

The collection

$$U + (\tau + 5)V + (\tau + 6)W$$

of $2\tau + 12$ cubes has no $a \times b \times c$ frame using all five colours. In particular, $U + 7V + 8W$ fails for both a $2 \times 3 \times 3$ frame and a $2 \times 2 \times 4$ box.

Proof. The corner types are

U	$003, 004, 013, 024, 032, 041, 123, 142$
V	$113, 114, 013, 024, 032, 041, 123, 142$
W	$133, 334, 012, 024, 031, 043, 132, 234.$

Triples denote cyclic classes. The only two-colour types in their union are $003, 004$, $113, 114$, and $133, 334$. An adjacent-repeat target must therefore repeat colour 0, 1, or 3. Its two repeated-colour corners force the colours on the opposite faces at the ends of its monochromatic edge to be, respectively, $\{3, 4\}$, $\{3, 4\}$, or $\{1, 4\}$. The two orders of the remaining colours give U, V, W, respectively, and their mirrors. The mirrors of U and V require 023, which is absent from the union; the mirror of W requires 021, also absent.

An opposite-repeat target requires both cyclic orientations of four distinct unordered tri-coloured triples. The union above contains both orientations only for 013 and 123. Thus no opposite-repeat target is possible. The only corner-supported targets are U, V, W.

Target U needs two copies possessing a 00 corner, but the collection has only one such copy.

For target V, seven of its corner types are absent from W; the only shared type is 024. In addition, exactly one geometric edge in each coordinate direction has a type absent from W: these are $14, 14, 11$. Consequently, in any orientation of an $a \times b \times c$ frame, $7 + \tau$ positions require a U or V source. There are only $\tau + 6$ such copies.

For target W, seven corner types are absent from both U and V. Again, one geometric edge in each direction has a private type, now $34, 33, 34$. Hence $7 + \tau$ positions require a W source, whereas only $\tau + 6$ are available.

Every possible target therefore fails (Figure 3). □

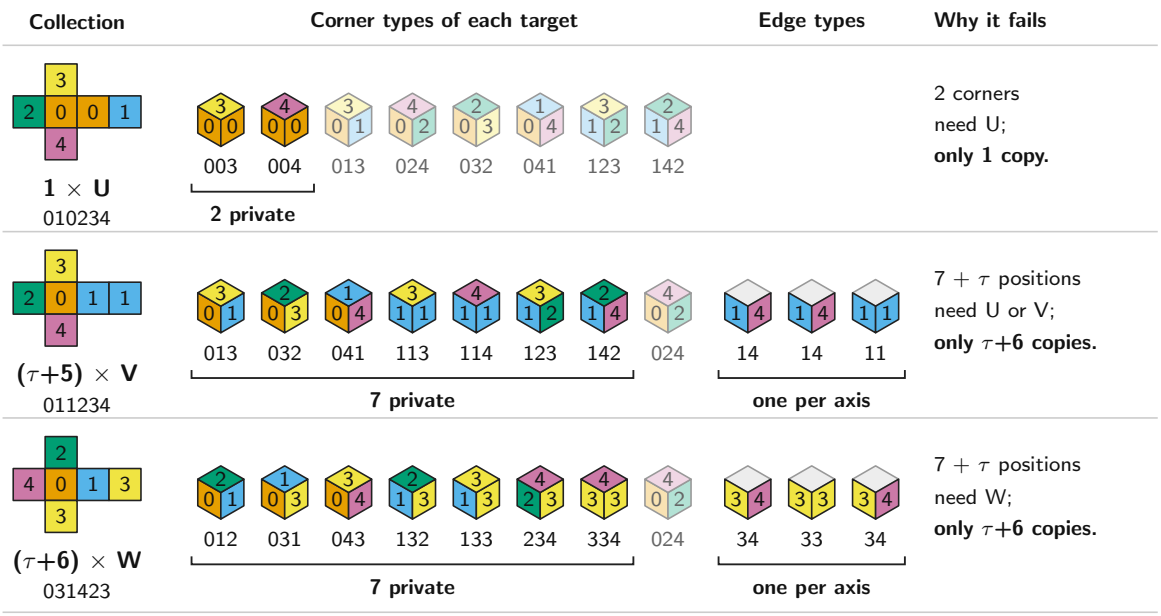


Figure 3: The uniform obstruction $U + (\tau + 5)V + (\tau + 6)W$, where $\tau = a + b + c - 6$. Each net has $+x$ at its centre, and corner types are read counterclockwise from outside. Only U, V and W can be targets. Their pools are U alone, U or V, and W alone; faded corner types also occur on a cube outside the pool. For V and W, one edge in each direction has a private type, and these edges have τ noncorner positions in total.

Proposition 19. *For every $a, b, c \geq 2$,*

$$F_5(a, b, c) \geq \max\{15,\ 4(a + b + c) - 16,\ 2(a + b + c) + 1\}.$$

Proof. The three terms follow from the fourteen-cube corner obstruction (Lemma 5), the number of frame positions, and Lemma 18, respectively. □

7.4 The smallest thin box

Theorem 20 (The $2 \times 2 \times 3$ box)**.** *We have*

$$F_5(2, 2, 3) = 15.$$

In particular, any fifteen five-colour cubes contain twelve that form a $2 \times 2 \times 3$ box with monochromatic exterior faces using all five colours.

Proof. The fourteen-cube corner obstruction (Lemma 5) gives the lower bound: the eight corner cubes of any box would themselves give a corner solution.

For the upper bound, take fifteen arbitrary source copies. Call a bichromatic pair rare if at most seven sources possess it. Each rare pair is missing from at least eight sources. The missing sets for any two rare pairs intersect, since $8 + 8 > 15$; hence the two pairs must be disjoint. There are therefore at most two rare pairs.

Suppose first that there is no rare pair. The corner theorem supplies a target T and a corner matching. If T is adjacent-repeat, choose the long axis perpendicular to its monochromatic edge. If T is opposite-repeat, any choice of long axis is suitable. In either case all four noncorner frame positions have bichromatic requirements, and each requirement has at least eight source neighbours.

If there is a rare pair $e = \alpha\beta$, at least eight sources belong to $\mathcal{H}_{\alpha\beta}$. Choose eight of them and any three additional copies. Lemma 17 gives a corner matching with target $T \in \mathcal{H}_{\alpha\beta}$. Orient

its $\alpha\beta$ axis along one of the two short directions, and choose a perpendicular direction as the long axis. The four noncorner edge requirements then each join a pole colour, α or β, to a ring colour. None is $\alpha\beta$, and none can be the other rare pair, if present, since that pair is disjoint from $\alpha\beta$ and uses only ring colours. The target's monochromatic edge, when present, is parallel to the $\alpha\beta$ axis and contributes no noncorner position. Again all four edge requirements are bichromatic and have at least eight source neighbours.

For either chosen target and orientation, every source fits at least four of the twelve positions, by Lemma 16. Thus each source misses at most eight positions.

Let D be any subset of the twelve target positions. If D contains only corners, its Hall inequality follows from the corner matching. Otherwise, if $|D| \le 8$, one of its edge positions has at least eight neighbours. If $|D| > 8$, every source neighbours D, because no source misses more than eight positions. Hall's theorem supplies a matching of all twelve positions to distinct source copies. The final matching may rearrange the preliminary corner placements. □

7.5 Sharp thresholds for $2 \times 3 \times c$ frames

Lemma 21 (Compatibility for $2\times3\times c$)**.** *Let $M = c-2 \ge 2$ and put $d = 2M+6$. Orient a five-colour target for a $2\times3\times c$ frame so that its monochromatic edge, if any, is parallel to the side of length two. An adjacent-repeat source misses at most d positions, and an opposite-repeat source misses at most $d+1$ positions. If a source misses $d+1$ positions, it is the unique opposite-repeat variety lacking the two bichromatic types appearing among those missed edge positions.*

Proof. The nonzero edge lengths are $1, M$. By Lemma 12, for an opposite-repeat source, at most six corners are incompatible. Its two missing bichromatic pairs are disjoint, so at most three geometric edges are incompatible, with at most two parallel to either nonzero direction. If their counts are k_1, k_M, the defect is at most

$$6 + k_1 + Mk_M \le 6 + 3 + 2(M-1) = 2M + 7 = d + 1.$$

Equality requires six incompatible corners, two incompatible long edges, and one incompatible unit-length edge. These three geometric edges display exactly two bichromatic types: there are at least two because any target type occurs at most twice, and at most two because the source lacks only two types. They are disjoint and determine the opposite-repeat source variety uniquely.

An adjacent-repeat source lacks only one bichromatic type. Against an opposite-repeat target, at most six corners and edge positions of total weight $2M$ are incompatible, giving defect at most d. Against an adjacent-repeat target, an incompatible edge lies transverse to the monochromatic edge, so Lemma 13 gives a shared corner. The absent type contributes weight at most $M+1$: a doubled type has one edge in each nonzero direction. The defect is therefore at most

$$7 + (M+1) = M + 8 \le 2M + 6 = d.$$

If no actual edge is incompatible, the defect is at most eight, which is also at most d. □

Lemma 22 (An opposite-repeat target along the short axis)**.** *Let $c \ge 3$. For a $2\times3\times c$ frame, orient an opposite-repeat target with its repeated-colour axis along the side of length two. Every source misses at most $2(c-2)+6$ positions.*

Proof. Every actual edge type contains the target's repeated colour. By Lemma 12, a source can lack at most one such bichromatic type, because its missing pairs are disjoint. The absent type accounts for at most two parallel geometric edges, of total weight at most $2(c-2)$. At least two corners are shared, so at most six further positions are missed. □

Theorem 23 (The remaining $2\times3\times c$ frames)**.**

$$F_5(2,3,3) = 17, \qquad F_5(2,3,c) = 4c+4 \quad (c \ge 4).$$

Proof. First let $c = 3$ and take seventeen sources. Call a bichromatic pair rare if at most eight sources possess it. Each rare pair is absent from at least nine sources, so any two rare pairs are disjoint by Lemma 12 and there are at most two. If no pair is rare, choose a corner target using $N_5 = 15$ and put its possible monochromatic edge along the side of length two. Otherwise choose a rare pair $\alpha\beta$ and apply Lemma 17 to eight sources in $\mathcal{H}_{\alpha\beta}$ and any three others. Orient the target's $\alpha\beta$ axis along the side of length two. All actual edge types join a pole colour to a ring colour, so they avoid both possible rare pairs and are bichromatic.

Every source misses at most nine of the sixteen frame positions. Indeed, if source or target is opposite-repeat, at most six corners and three actual geometric edges are incompatible. If both are adjacent-repeat, at most two actual geometric edges are incompatible; whenever one is incompatible, Lemma 13 supplies a shared corner. If no edge is incompatible, at most eight positions are missed.

The chosen target has a corner matching, and every edge requirement has at least nine neighbours. Lemma 11, with the corners as the matched subset, completes the frame. This proves $F_5(2, 3, 3) \le 17$; the collection $U + 7V + 8W$ in Lemma 18 gives equality.

Now let $c \ge 4$, put $M = c - 2 \ge 2$, and set

$$d = 2M + 6, \qquad N = 4M + 12 = 2d.$$

There are $N = 4c + 4$ frame positions. Take any N source copies, and call a bichromatic pair rare if at most $d-1$ copies possess it. Each rare pair is absent from at least $d+1 > N/2$ sources. Consequently any two rare pairs are disjoint, and there are at most two.

If no pair is rare, use the corner theorem and orient the target's possible monochromatic edge along the side of length two. Otherwise, choose a rare pair $\alpha\beta$. At least $d+1 \ge 11$ sources belong to $\mathcal{H}_{\alpha\beta}$. Lemma 17 supplies a corner solution whose target belongs to that family. Orient its $\alpha\beta$ axis along the side of length two. All actual edge types join a pole colour to a ring colour, and hence avoid both rare pairs; the monochromatic edge, if present, has no noncorner positions.

In either case the chosen target has a corner matching and every actual edge requirement has at least d source neighbours. Hall's inequality holds for corner-only sets, and for any set of at most d positions containing an edge. By Lemma 21, sets of more than $d + 1$ positions neighbour every source.

Suppose the frame has no matching. A deficient Hall set D must therefore have exactly $d + 1$ positions. Every nonneighbour misses all of those positions, so Lemma 21 says that all nonneighbours have one opposite-repeat variety V, determined by the two absent bichromatic types. Since D has at most d neighbours among $N = 2d$ copies, the inventory contains at least d copies of V.

Retarget the frame to V with its repeated-colour axis along the side of length two. By Lemma 22, every source misses at most d positions. The d copies of V are compatible with every position, so they verify Hall's inequality for every set of size at most d. Larger sets neighbour every source. This gives a frame after all.

Thus N sources always suffice. The number of frame positions supplies the matching lower bound. □

7.6 Sharp thresholds for $2 \times 2 \times c$ boxes

Put $M = c-2$. The frame of a $2\times2\times c$ box consists of eight corners and four parallel geometric edges, each with M positions. Its size is $L = 4M + 8 = 4c$. We call an orientation of an adjacent-repeat target *transverse* if its monochromatic edge is not parallel to the long axis. For an opposite-repeat target, its *axial* orientation has the repeated faces perpendicular to the long axis.

Lemma 24 (Directional compatibility)**.** *Let $M \ge 2$.*

1. *For a transverse adjacent-repeat target, every adjacent-repeat source misses at most $M+7$ positions. Every opposite-repeat source misses at most $2M+6$ positions. An opposite-repeat source missing more than $M+6$ positions must be one of two varieties; their missing long-edge types are complementary pairs.*

2. *For an axial opposite-repeat target V, every source misses at most $d = 2M+4 = L/2$ positions. If $M \geq 3$, a source different from V can miss d positions only if it is one of the other two opposite-repeat varieties with the same repeated colour. These two varieties have complementary missing long-edge types.*

Proof. For (i), write the target in face order $(+x, -x, +y, -y, +z, -z)$ as

$$(r, \alpha, r, \beta, \gamma, \delta),$$

and take the long direction to be x. Its four long-edge types are

$$r\gamma, \quad r\delta, \quad \beta\gamma, \quad \beta\delta.$$

An adjacent-repeat source lacks at most one of these. If it lacks one, Lemma 13 gives a shared corner, so its defect is at most $M+7$. If it lacks none, its defect is at most eight, also bounded by $M+7$.

By Lemma 12, an opposite-repeat source shares at least two corners and lacks at most two long-edge types, giving defect at most $2M+6$. To lack two of the four displayed types, its two missing pairs must be one of the disjoint matchings

$$\{r\gamma, \beta\delta\} \qquad \text{or} \qquad \{r\delta, \beta\gamma\}.$$

Each matching determines exactly one opposite-repeat variety, with repeated colour α. The two varieties have complementary missing long-edge types. Every other opposite-repeat source lacks at most one long-edge type, and hence has defect at most $M+6$.

For (ii), let the repeated colour of V be r, and put its singleton opposite pairs on the two transverse axes. The four long-edge types are the four edges of the resulting $K_{2,2}$ on the singleton colours. The other two opposite-repeat varieties with repeated colour r each share four corners and two long-edge types with V. Their compatible position count is $4+2M = d$, and their missing long-edge pairs are complementary.

Every adjacent-repeat source lacks at most one long-edge type. An opposite-repeat source with repeated colour different from r also lacks at most one: one of its two missing pairs contains r, whereas no long-edge type does. By Lemma 12, these sources share at least two corners with V, so they fit at least $2+3M \geq d$ positions. If $M \geq 3$, this last inequality is strict. The target variety V itself fits every position. This proves both assertions. □

Lemma 25 (Concentrated opposite-repeat copies)**.** *Let $M \geq 3$ and $d = 2M+4$. A collection of $2d = 4M+8$ cubes containing at least $d-1$ copies of one opposite-repeat variety forms a $2 \times 2 \times (M+2)$ box.*

Proof. Let that variety be V with multiplicity v. Use its axial target orientation. Every source misses at most d positions. If $v \geq d$, Hall's inequality is immediate: sets of at most d positions have the v copies of V as neighbours, and larger sets neighbour every source.

Suppose $v = d-1$. The same reasoning handles all position sets except sets D of size exactly d. If such a set fails Hall, no non-V copy neighbours it. Those copies must each miss d positions, so Lemma 24(ii) restricts them to the other two opposite-repeat varieties with the same repeated colour. Since $d > 8$, D contains a long-edge position. The two candidate varieties have complementary missing long-edge types, so at most one can be absent from the neighbourhood of D. All $2d - v = d+1$ non-V copies are therefore one variety W. Using W as the axial target gives the previous case, with at least d universal copies, and completes the box. □

Lemma 26 (A target avoiding scarce long-edge types)**.** *Suppose a collection has $N \geq 15$ cubes. Call a bichromatic pair rare when at most ρ sources possess it, where $N-\rho \geq 8$ and $2(N-\rho) > N$. Then there is a corner-solvable target, transverse if adjacent-repeat and axial if opposite-repeat, whose four long-edge types are all non-rare.*

Proof. Each rare pair is missing from more than half the collection. Thus any two rare pairs are disjoint by Lemma 12, and there are at most two. If there is no rare pair, use the corner theorem and choose either specified orientation.

Otherwise let $\alpha\beta$ be rare. At least eight sources belong to $\mathcal{H}_{\alpha\beta}$. Lemma 17 supplies a corner target in this family. Its $\alpha\beta$ axis is a singleton opposite pair and can be put along a short direction. If the target is opposite-repeat, choose its repeated-colour axis as the long direction; otherwise choose either axis perpendicular to the $\alpha\beta$ axis. Every long-edge type then joins one of α, β to a ring colour. It is neither $\alpha\beta$ nor the other possible rare pair, whose colours are both ring colours. These orientations are axial or transverse as required. □

Theorem 27 (Thin rods)**.**

$$F_5(2,2,4) \leq 17, \qquad F_5(2,2,c) = 4c \quad (c \geq 5).$$

Proof. First let $c = 4$, so $M = 2$ and $L = 16$, and take seventeen source copies. In Lemma 26 use $\rho = 8$. The chosen target has a corner matching, and every long-edge type has at least nine sources. If the target is opposite-repeat, its defect is at most eight by Lemma 24, and Hall's theorem completes it.

For an adjacent-repeat target, adjacent-repeat sources miss at most nine positions and opposite-repeat sources at most ten. A failing Hall set D cannot consist only of corners. Since it contains an edge position, it has at least nine neighbours; and if $|D| > 10$, it neighbours every source. Therefore a failure would have $|D| = 10$ and exactly nine neighbours. Its eight non-neighbouring source copies must all be opposite-repeat. Each must miss two long-edge types, since otherwise its defect is at most $M + 6 = 8$. The two candidate varieties in Lemma 24(i) have complementary missing long-edge types. As D contains an edge position, all eight copies must be the same opposite-repeat variety V.

The axial target V is solved by Hall directly: its eight copies neighbour every nonempty set of at most eight positions, and larger sets neighbour all seventeen sources, since each misses at most eight positions. This proves the seventeen-cube upper bound.

Now let $M = c-2 \geq 3$, put $d = 2M+4$, and take exactly $N = 2d = L$ cubes. Use $\rho = d-1$ in Lemma 26. Every long-edge type of the chosen target has at least d sources. An opposite-repeat target is completed directly by Hall, since every source misses at most d positions.

For an adjacent-repeat target, adjacent-repeat sources miss at most $M + 7 \leq d$ positions; opposite-repeat sources miss at most $2M + 6 = d + 2$. Any failing Hall set D therefore has

$$d + 1 \leq |D| \leq d + 2.$$

It has at most $d + 1$ neighbours, leaving at least $d - 1$ non-neighbouring source copies. All are opposite-repeat. An opposite-repeat source missing only one long-edge type has defect at most $M+6 < d+1$, so all these copies are among the two varieties in Lemma 24(i). Since D contains an edge, all must be one variety V. Lemma 25 then supplies a box from the original collection. Thus every collection of $L = 4c$ cubes works. The number of positions gives the matching lower bound. □

Proposition 28 (Complete rod threshold)**.** *Together with Theorem 3, Theorem 20 and the sixteen-cube obstruction of Lemma 18, Theorem 27 gives*

$$F_5(2,2,c) = \max\{15,\ 4c,\ 2c+9\} \qquad (c \geq 2).$$

Equivalently, the values are $15, 15, 17$ for $c = 2, 3, 4$, respectively, and $4c$ for every $c \geq 5$.

Proof. The cases $c = 2, 3$ are the established corner and smallest thin-box thresholds. For $c = 4$, Lemma 18 and the seventeen-cube upper bound give equality. For $c \geq 5$, the number of positions gives $F_5(2, 2, c) = 4c$, as proved above. □

7.7 Boxes whose two shortest sides sum to at least six

Order the side lengths as $2 \leq a \leq b \leq c$ and put

$$m = a - 2, \qquad t = b - 2, \qquad M = c - 2, \qquad \tau = m + t + M.$$

The frame has $L = 8 + 4\tau$ positions. In this section assume $m + t \geq 2$.

Lemma 29 (Weighted compatibility)**.** *Let $2 \leq a \leq b \leq c$ with $a + b \geq 6$, so that $m + t \geq 2$ and $M \geq 1$. Every source misses at most $d = \tau + M + 6$ positions of every target frame.*

Proof. By Lemma 12, if the target is opposite-repeat, at most three geometric edges are incompatible, with at most two in any direction. Their total weight is at most $2M + t \leq \tau + M$. At most six corners are incompatible, so the claimed bound follows.

For an opposite-repeat source, take the adjacent-repeat target

$$(r, \alpha, r, \beta, \gamma, \delta),$$

whose monochromatic edge is parallel to z. A doubled bichromatic type, $r\gamma$ or $r\delta$, occupies one edge in each of the x, y directions. If such a type is missing, it and the monochromatic edge contribute one edge in each direction; the other disjoint missing type contributes at most one further edge. The missed weight is at most $\tau + M$. If neither missing type is doubled, two missing single types cannot have the same direction: the single types in each direction share a colour. Adding the monochromatic edge therefore gives at most three edges, with at most two parallel, again of weight at most $2M + t \leq \tau + M$. There are at most six incompatible corners.

For two adjacent-repeat varieties, at most three geometric edges are incompatible. If there are at most two, the total defect is at most

$$2M + 8 \leq \tau + M + 6,$$

because $m + t \geq 2$. If there are three, they are the monochromatic edge and a doubled missing bichromatic type, one edge in each direction. Lemma 13 gives a shared corner. The defect is at most $\tau + 7 \leq \tau + M + 6$, since $M \geq 1$. □

Theorem 30. *If $2 \leq a \leq b \leq c$ and $a + b \geq 6$, then*

$$F_5(a, b, c) = 4(a + b + c) - 16.$$

Proof. Take $N = L = 8 + 4\tau$ source copies and put $d = \tau + M + 6$. Call a bichromatic pair rare if at most $d - 1$ copies possess it. Its missing family has at least

$$N - d + 1 = 3\tau - M + 3$$

copies, exceeding $N/2$ by $m + t - 1 \geq 1$. Thus any two rare pairs are disjoint by Lemma 12, and there are at most two.

By Lemma 11, a target avoiding all rare pairs has a full frame matching whenever its corners and monochromatic edge positions have a matching: every other position has at least d neighbours, and Lemma 29 bounds each source's defect by d.

No rare pair. For each colour reserve up to m sources having that colour on adjacent faces, taking all such sources if fewer exist. The reservations are disjoint. At least $N - 5m \geq 7m + 8 \geq 15$ sources remain when $m \geq 1$; when $m = 0$, none are reserved and $N \geq 24$. The corner theorem supplies a target on the unreserved sources. Orient its possible monochromatic edge along the

shortest direction. If the target repeats colour r adjacently, a compatible source with adjacent r-faces survived the reservation, so the r-reserve has its full size m. It fills the monochromatic edge. The Hall observation completes the frame.

One rare pair $\alpha\beta$. For each of the three ring colours, reserve up to m sources in the whole collection having that colour on adjacent faces, again taking all such sources if fewer exist. At most $3m$ family copies are removed, so $\mathcal{H}_{\alpha\beta}$ retains at least

$$(3\tau - M + 3) - 3m = 3t + 2M + 3 \geq 8$$

copies, and at least eleven total sources remain unreserved. Apply Lemma 17 to eight unreserved family copies and any three further unreserved sources. The resulting corner target lies in $\mathcal{H}_{\alpha\beta}$. Orient its $\alpha\beta$ axis along the shortest direction. Any monochromatic edge is parallel to this axis and has its full reserve of m copies, by the same survivor argument. The target avoids the only rare pair, so the Hall observation applies.

Two rare pairs e, f. Their common missing variety V is the unique opposite-repeat variety with e, f as singleton opposite pairs. Its multiplicity satisfies

$$v \geq N - 2(d-1) = 2(m+t) - 2 \geq 2.$$

If $v \geq 8$, use eight copies for the corners of V. If $4 \leq v \leq 7$, the family $\mathcal{H}_e$ has at least eleven copies and hence at least four not of variety V. Each shares at least four corners with V, by the ring array. Place four such copies at distinct corners and fill the other four with V. In either case, V has no monochromatic edge and avoids both rare pairs; the Hall observation applies.

It remains that $v = 2$ or 3, forcing $m + t = 2$ and $N = 2d$. If n_e, n_f count the sources possessing the rare pairs, then

$$n_e, n_f \leq d - 1, \qquad v \geq N - n_e - n_f.$$

Consequently,

$$n_e, n_f \geq d - 2, \qquad n_f + v \geq d + 1.$$

Use exactly the reservation and the construction of Lemma 17 from the one-rare-pair case to obtain a corner-and-monochromatic-edge matching for a target $T \in \mathcal{H}_e$, with its e axis shortest.

Within this family, V shares at least four target corners and can miss at most two geometric edges, both parallel to the e axis. For a different opposite-repeat target these are its two f-edges; for an adjacent-repeat target they are its f-edge and monochromatic edge. Thus V misses at most $2m + 4 \leq 6$ actual positions.

The only Hall sets not covered as before contain bichromatic edge positions of type f and no non-rare bichromatic edge. Such a set D has at least n_f neighbours. If $|D| > n_f \geq d - 2 \geq 8$, every copy of V is also a neighbour, since it misses at most six positions. These are distinct from the n_f copies possessing f, so

$$|\Gamma(D)| \geq n_f + v \geq d + 1.$$

This handles $|D| \leq d$; larger sets neighbour every source. Hall's theorem completes the frame in the final case.

The number of frame positions gives the lower bound, proving equality. All preliminary placements certify Hall inequalities; the final matching may rearrange them. □

7.8 Completion of the classification

Proof of Theorem 1. Order the sides as $2 \leq a \leq b \leq c$. When $a + b \geq 6$, Theorem 30 gives $F_5 = L$. The remaining possibilities are exactly $(a, b) = (2, 2)$ and $(2, 3)$, settled by Proposition 28 and Theorem 23. Together with the corner theorem at $(2, 2, 2)$, these give $F_5 = 15$ for sums

six and seven, $F_5 = 17$ for sum eight, and $F_5 = L$ for every larger sum. Proposition 19 proves sharpness and yields the stated maximum formula.

For the complete box, take $\max\{abc, F_5(a,b,c)\}$ sources and first choose a frame. Every source has every colour, so any unused source can fill any face-interior position after a suitable proper rotation. All fully interior positions are unrestricted. There are enough unused sources to fill the remaining $abc - L$ positions. Conversely, any complete box supplies a frame and uses abc sources. Hence

$$G_5(a,b,c) = \max\{abc, F_5(a,b,c)\}.$$

Putting $x = a-2$, $y = b-2$, and $z = c-2$ gives

$$abc - L = xyz + 2(xy + xz + yz) \geq 0,$$

which simplifies this identity to the claimed formula for G_5.

If $(a,b) = (2,2)$, this formula is $\max\{4c, 15, 2c+9\}$: it exceeds the volume only for $c = 2,3,4$. In every other sorted case, $b \geq 3$; then $abc \geq 18 \geq 15$ and

$$abc - 2(a+b+c) - 1 \geq 2bc - 2b - 2c - 5 \geq 1.$$

The first inequality follows by increasing a from two, and the second is minimised at $b = c = 3$. There are no further exceptions. □

8 Constructing an assembly

The threshold theorem guarantees some target colouring, which need not be specified in advance. It can be found by a bounded number of capacitated matching problems, even when the side lengths are large.

Proposition 31 (Compressed construction)**.** *Given an explicit list of q sources and $a,b,c \geq 2$, one can find a frame, or report that none exists, in $O(q+L)$ time, where $L = 4(a+b+c) - 16$. A complete-box assembly can be found, or ruled out, in $O(q + abc)$ time. These bounds use the word-RAM model with the input sizes, dimensions and output indices fitting in a word. On collections covered by the corresponding threshold, the running time is $O(q)$. For sources given as 75 binary multiplicities, feasibility and a compressed assembly have polynomial bit complexity; expanding the placements costs the output size.*

Proof. Classify each source among the 75 varieties by testing its 24 proper rotations, recording both its variety and a rotation to the representative. This takes $O(q)$ time and gives 75 multiplicities and lists of actual copies.

Test each target variety with each distinct assignment of the three side lengths to its coordinate axes. There are at most six assignments, hence at most 450 configurations. When exactly two side lengths agree, three assignments suffice, giving 225 configurations; when all three agree, only 75 are needed. Every physical target is a proper rotation of one of the 75 representatives, and every proper rotation permutes the three coordinate axes. Bringing the target to its representative therefore permutes the side-length assignment. Sign changes only reverse positions along an edge; they introduce no additional demand pattern. This proves that the listed configurations cover every assembly.

For each configuration, use a flow network with 75 source-variety nodes and at most twenty requirement nodes: eight corners of demand one and twelve geometric edges whose demands are their assigned side lengths minus two. Omit zero demands. Add an arc from the network source to each variety with capacity its multiplicity, and an arc from each requirement to the sink with capacity its demand. Connect a variety to a requirement exactly when it fits, with capacity q. There are at most 97 vertices, including source and sink. Identical requirements may be merged.

An integral flow of value L saturates every requirement and gives the number of copies of each variety to place there. Conversely, any frame induces such a flow. A shortest-augmenting-path max-flow algorithm has a number of arithmetic operations bounded in terms of the network size, independently of the capacities [4]. Here the network size and the number of configurations are constants. Thus all flow tests take $O(1)$ word operations, or polynomial time in the bit lengths for binary input. If every test fails, no frame exists.

Expand a successful flow into L distinct source copies and positions. A precomputed proper rotation for each compatible corner or edge, composed with the recorded source rotation, gives each placement's orientation. This takes $O(L)$ time. A complete box exists exactly when a frame exists and $q \geq abc$: fill face interiors with unused copies rotated to expose the required single colour, and fill the interior arbitrarily. Writing these placements takes $O(abc)$ time. □

9 Collections with few varieties

Let $M(k)$ be the least number of cubes guaranteeing a corner solution when the collection contains at most k varieties.

Lemma 32. *Two five-colour varieties with disjoint corner sets can build an opposite-repeat target from four copies of each.*

Proof. This is a small finite geometric check. Of the $\binom{75}{2}$ unordered pairs, 330 have disjoint corner sets. Up to colour permutation and exchange of the sources, they have the following five representatives. The first source is $(0, 1, 0, 2, 3, 4)$ throughout; tuples use the face order of Section 2. Each listed target has four corners supplied by each source.

Second source	Target
$(0, 1, 1, 2, 4, 3)$	$(0, 0, 1, 2, 3, 4)$
$(0, 1, 2, 3, 2, 4)$	$(0, 1, 2, 2, 3, 4)$
$(0, 1, 2, 3, 3, 4)$	$(0, 1, 2, 2, 3, 4)$
$(0, 1, 2, 4, 3, 4)$	$(0, 1, 2, 2, 3, 4)$
$(0, 3, 0, 4, 1, 2)$	$(0, 0, 1, 2, 3, 4)$

The accompanying program `src/five_colour_extensions.cpp` regenerates the varieties and checks the orbit coverage and all five witnesses. □

Theorem 33. *For every integer $k \geq 1$,*

$$M(k) = \begin{cases} 8 & k = 1, \\ 13 & k = 2, \\ 15 & k \geq 3. \end{cases}$$

Consequently, three is the minimum number of varieties in a fourteen-cube collection without a solution.

Proof. Eight identical copies build their own variety, and fewer than eight pieces cannot form a corner solution. For two varieties, consider thirteen copies. If either variety occurs at least eight times, it builds itself. Otherwise the multiplicities are seven and six. If the varieties share a corner type, use a minority copy at that corner of the majority target and seven majority copies elsewhere. If their corner sets are disjoint, apply Lemma 32.

For the lower bound take six copies each of

$$U = (0, 1, 0, 2, 3, 4), \qquad Q = (0, 1, 1, 3, 2, 4),$$

where U is the variety of Lemma 18. Their corner sets share just 041. Inspection of the 75 varieties shows that only U, Q have all corner types in their union. Each has seven private positions, so neither can be built with six copies. The same geometric checker verifies these assertions. Thus $M(2) = 13$. For $k \geq 3$, Theorem 3 supplies the upper bound, and the collection in Lemma 5 supplies the lower bound. □

Remark (A family of fourteen-cube obstructions). Keep A, C from Section 3, and put

$$B_1 = (0, 1, 2, 3, 2, 4), \quad B_2 = (0, 1, 2, 3, 3, 4), \quad B_3 = (0, 1, 2, 4, 3, 4).$$

All 21 collections

$$4A + 5C + x_1B_1 + x_2B_2 + x_3B_3, \qquad x_1 + x_2 + x_3 = 5, \quad x_i \in \mathbb{Z}_{\geq 0},$$

are unsolvable. The B_i share the six tricoloured corner types $023, 034, 042, 124, 132, 143$. The combined corner union allows only A, C, B_1, B_2, B_3 as targets, as a check of the catalogue confirms. Target A needs five A-copies, C needs six C-copies, and any B_i needs six copies from the B-group. The respective supplies are four, five and five, regardless of their distribution within the B-group. The original example is $x_1 = 5, x_2 = x_3 = 0$.

10 Formalisation and reproducibility

The development uses Lean 4 and mathlib [8, 9], both at `v4.35.0`. It proves the physical theorems, including the connection between cube arrangements and the finite certificate problems. The physical model and palette-independent geometry come from the shared `colour-cubes-core` library [6].

Lean proves completeness and uniqueness of the 75 varieties and checks all 45,000 corner-compatibility bits using the shared corner–rotation theorem. The lower bound uses 75 deficient corner sets: 72 singletons and three sets of sizes 5, 6 and 6. Its theorem is `FiveColour.Physical.lower_bound` in `FiveColour/LowerBound.lean`. This replaces the elementary target classification in Lemma 5; catalogue completeness and rotation transport extend the result to every physical target.

The upper-bound formalisation follows the Hall reduction, symmetry transport, anchor selection and shared-corner argument. Finite checks verify capacities, cover representatives and the reduction from 357 roots to 279. The deletion step is `FiveColour.Physical.root_projection` in `FiveColour/RootProjection.lean`. The bridge to the archive, in `FiveColour/ProjectedBridge.lean`, identifies the full admissibility predicate: coordinate bounds, total, budget and all Hall-cut families.

For efficient replay, Lean proves soundness of a packed implementation of the interval checker. For multiplicities $0 \leq x_i \leq 7$, index the varieties by $0, \ldots, 74$ and put

$$P(x) = \sum_{i=0}^{74} x_i B^i, \qquad W(S) = \sum_{i \in S} B^{74-i}, \qquad B = 1024.$$

Then

$$x(S) = \left\lfloor \frac{P(x)W(S)}{B^{74}} \right\rfloor \bmod B.$$

Every coefficient in the product is at most $75 \cdot 7 < 1024$, so no carry can affect this coefficient extraction. Lean proves the identity, decoding, coordinate updates and equivalence with the integer inequalities, and checks all 19,125 neighbourhood weights.

The 279 certificates are split into 4,786 chunks of at most 128 instructions. Composition requires exact boundary-vector agreement and a proof for every referenced box; the exporter is untrusted. All chunks and their final assembly have passed kernel checking. The

case mapping and certificate hashes are in `data/q5-projected-root-manifest.json`. The generated module `FiveColour/ProjectedChunks/Threshold.lean` proves the unconditional theorems `threshold15` and `sharpThreshold15`. Both lie in namespace `FiveColour.Packed.ProjectedChunks`.

The cuboid proof in Lean follows Lemma 17. The cube-only exports, such as `frame20`, retain an older proof through an eleven-cube missing-pair lemma in `FiveColour/RingEleven.lean`: its nine-variety pattern and ten Hall cuts per target are kernel-checked, and the multiplicity argument is proved without enumerating collections. Lean's frame-completion step adapts the general construction of [5]; no theorem of that paper is used.

The small-support proofs check pair witnesses directly, without reducing them to five colour orbits, and use deficient corner sets for the two obstruction constructions. The modules are `SmallSupport.lean` and `MixedObstructions.lean` in the `FiveColour` directory.

The full cuboid classification is formalised in `FiveColour/BoxThresholds.lean`. In namespace `FiveColour.Physical`, `box_frame_threshold_iff` and `box_threshold_iff` give necessary and sufficient pool sizes for arbitrary side lengths at least two, assuming `FiveColour.Physical.Threshold15`. `Solution.lean` applies the completed corner theorem to discharge this assumption. In namespace `FiveColour.Puzzle`, it exports the unconditional theorems `box_frame_threshold` and `complete_box_threshold`, together with `frame20`, `frame_threshold_all`, `full_cube`, `complete_cube_threshold` and `support_threshold`.

The rectangular model and completion are in `FiveColour/BoxModel.lean`; compatibility is connected to proper rotations, and axis assignments are realised by proper rotations. `FiveColour/EightPlusThree.lean` proves the physical prescribed-family lemma. Its proof uses a failed Hall set to obtain the small cover directly, as in Lemma 17; signed projection and small matching patterns are checked in the kernel. The weighted bounds and upper deductions are in `FiveColour/WideFrames.lean`, `FiveColour/RodFrames.lean` and `FiveColour/StripFrames.lean`. The thin-box proofs formalise both concentration and switching targets, including the equality cases of the affine geometric bounds. `FiveColour/BoxLowerBound.lean` proves the uniform three-variety obstruction for every shape and orientation. A separate C++ audit reconstructs the geometry and checks these profiles. No new interval certificates are used. The flow algorithm and its complexity bound in Section 8 are mathematical deductions, not additional Lean claims.

The dependency audits cover 133 declarations listed in `Audit.lean`, fourteen extension statements listed in `ExtensionsAudit.lean`, and the final replay theorems. Their dependencies are confined to `propext`, `Classical.choice` and `Quot.sound`. No `native_decide` is used. Reproduce the proof and audits with:

```
make check WORKERS=2
make projected-replay-data WORKERS=8
make projected-replay WORKERS=8
make check-extensions
```

The module guide is `FiveColour/README.md`.

A Mathlib-only comparator challenge states the corner problem geometrically, restating the library's definitions locally. It covers only the corner problem: the cuboid statements rest on the definitions in `FiveColour/BoxModel.lean`. Its equivalence with the development's model is proved by `FiveColour.Puzzle.hasSolution_iff` in `FiveColour/ComparatorBridge.lean`. The solution module applies the completed threshold theorem. A full external comparator pass remains unconfirmed; it is separate from the completed Lean proof.

11 Remaining questions

The sharp frame and complete-box thresholds leave the structure of extremal corner obstructions to be determined. Can all fourteen-cube obstructions be classified? The support bound and the

family in Section 9 give initial constraints and examples. A second question is whether $N_5 = 15$ admits a short elementary proof, replacing the finite certificates used here. Finally, boxes with a side of length one fall outside the model, since a source there exposes opposite faces; their thresholds remain open.

AI disclosure and review status

GPT-6-Astra (OpenAI), used through Codex, developed the arguments, search and verification programs, Lean formalisation and manuscript under the human author's direction. The author initiated and directed the research. Author review and submission approval of this working manuscript are pending. The inherited geometry has its own provenance in the pinned dependency. Automated checks and AI review are not independent human peer review.

References

[1] P. A. MacMahon, *New Mathematical Pastimes*, Cambridge University Press (1921). Digitised edition.

[2] E. Berkove, A. Eickemeyer and M. Schulman, *The 4-Color Cubes Puzzle*, Integers **18** (2018), A68. doi:10.5281/zenodo.10680216. Journal PDF.

[3] P. Hall, *On Representatives of Subsets*, Journal of the London Mathematical Society **s1-10**(1) (1935), 26–30. doi:10.1112/jlms/s1-10.37.26.

[4] J. Edmonds and R. M. Karp, *Theoretical Improvements in Algorithmic Efficiency for Network Flow Problems*, Journal of the ACM **19**(2) (1972), 248–264. doi:10.1145/321694.321699.

[5] J. Davies, *Three-colour solutions from four-colour cubes*, manuscript (2026). Accompanying repository.

[6] J. Davies, `colour-cubes-core`: palette-independent Lean definitions and proofs for face-coloured cubes, Lean library (2026). Release `v1.0.0`, `https://github.com/jasondavies/colour-cubes-core`.

[7] M. Rosenfeld, *MacMahon Cubes and Moshe's Insanity, Puzzle 1*, Geombinatorics Quarterly **36**(2) (October 2026), starts at p. 78. Journal listing; full text not examined.

[8] L. de Moura and S. Ullrich, *The Lean 4 Theorem Prover and Programming Language*, in Automated Deduction—CADE 28, LNCS **12699**, Springer (2021), 625–635. doi:10.1007/978-3-030-79876-5_37.

[9] The mathlib Community, *The Lean Mathematical Library*, in Proceedings of CPP 2020, ACM (2020), 367–381. doi:10.1145/3372885.3373824. arXiv:1910.09336.